

Hybrid Dynamic Programming and Divide-and-Conquer Strategies for DAG Task Scheduling on Heterogeneous Cloud Resources: A Comparative C++ Study

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Abstract—Scheduling computational workflows represented as Directed Acyclic Graphs (DAGs) onto heterogeneous Virtual Machines (VMs) is a fundamental NP-hard problem in cloud computing. This paper presents and compares five scheduling algorithms implemented in pure C++17 in the following order: (1) *HEFT* as the classical baseline, (2) *DP-HEFT* with bilateral-rank and 2-task look-ahead, (3) *Enhanced DP (EDP-HEFT)* with global DP assignment, (4) *Divide-and-Conquer (D&C)* with level decomposition, and (5) *Merge*, a hybrid scheduler that combines adaptive strategy selection with refinement. All algorithms share a unified communication-cost model and a reproducible DAG generator. Comparative analysis highlights the complexity-quality trade-off between fast greedy scheduling and more expensive global/refined optimisation.

Index Terms—DAG scheduling, cloud computing, HEFT, dynamic programming, divide and conquer, heterogeneous computing, makespan minimisation, bilateral rank, C++, workflow scheduling

I. INTRODUCTION

Cloud computing platforms increasingly execute scientific and enterprise workflows as Directed Acyclic Graphs (DAGs), where each node is a computational task and each directed edge encodes a data dependency [1], [2]. The scheduling problem — assigning tasks to heterogeneous Virtual Machines (VMs) while minimising total completion time (makespan) — is NP-hard in the general case [3], [4], motivating a rich body of heuristic research.

The seminal Heterogeneous Earliest Finish Time (HEFT) algorithm [5] achieves near-optimal makespan through upward-rank prioritisation combined with an Earliest Finish Time (EFT) insertion policy. While HEFT is efficient and well-studied, its greedy VM selection ignores downstream tasks, occasionally leaving critical successors stranded on slow resources. Extending HEFT with dynamic programming look-ahead is therefore a natural direction for improving solution quality.

This paper makes the following contributions:

- 1) **HEFT Baseline**: A faithful greedy HEFT implementation used as the classical reference for makespan and complexity comparison in the same C++17 environment.
- 2) **DP-HEFT**: A HEFT extension that uses bilateral rank (upward rank + downward rank) for task prioritisation and a weighted 2-task look-ahead DP for VM assignment, scoring each candidate as $EFT(t_i) + 0.25 \cdot EFT^*(t_{i+1})$.
- 3) **EDP-HEFT**: An enhanced variant that constructs a full $n \times m$ DP table over the bilateral-rank priority order, propagating scheduling snapshots to achieve globally optimal VM assignments within that order, and backtracking to recover the optimal sequence.
- 4) **D&C Scheduler**: A divide-and-conquer approach that decomposes the DAG into independent topological lev-

els via BFS, schedules each level as a sub-problem, and merges partial schedules with cross-level dependency and communication cost awareness.

- 5) **Merge Scheduler:** A hybrid merge strategy that adaptively switches among D&C, DP look-ahead, and EDP-style global assignment, followed by refinement passes to reduce bottlenecks and improve final makespan.
- 6) A shared C++17 implementation including a probabilistic DAG generator with reproducible fixed seeding, an exhaustive 8-check validator, and a unified output model enabling direct metric comparison.

The remainder of the paper is organised as follows. Section II surveys related work. Section III formalises the scheduling problem. Section IV describes the system architecture. Section V presents each algorithm with pseudocode. Section VI details the mathematical model. Section VII analyses computational complexity. Section VIII discusses C++ implementation. Section IX describes the experimental setup. Section X presents performance results. Section XI discusses findings. Section XII outlines future work. Section XIII concludes.

II. RELATED WORK

A. List Scheduling and HEFT

Kwok and Ahmad [6] provided a comprehensive taxonomy of static DAG scheduling algorithms, establishing upward rank as the dominant prioritisation criterion. Topcuoglu *et al.* [5] introduced HEFT and CPOP, demonstrating that upward-rank sorting with EFT insertion consistently achieves lower makespan than random or FIFO ordering on heterogeneous systems. Arabnejad and Barbosa [7] extended this work with an Optimistic Cost Table (OCT) that approximates future communication costs, improving makespan by up to 12% on standard benchmarks. Hagra and Janeček [8] proposed HBMCT, which combines task duplication with heterogeneity-aware priority.

B. Look-Ahead and DP Extensions

Bittencourt *et al.* [9] introduced HEFT-NC (No Communication), demonstrating that look-ahead awareness of immediate successors yields measurable makespan reductions. Sakellariou and Zhao [10] proposed a hybrid scheduler combining list scheduling with task duplication, exploiting idle processor windows. Bajaj and Agrawal [11] formalised the bilateral rank ($\text{rank}_u + \text{rank}_d$) and showed it captures both path-criticality and positional importance, outperforming pure upward-rank ordering.

C. Divide-and-Conquer Scheduling

Dong *et al.* [12] applied level-based decomposition to Grid scheduling, independently optimising intra-level task placement. Canon *et al.* [13] benchmarked decomposition heuristics against list schedulers on standard DAG suites, finding that level-based methods perform competitively on shallow graphs but degrade on deep critical paths.

D. Cloud Workflow Scheduling

Pandey *et al.* [14] adapted Particle Swarm Optimisation for cloud workflow scheduling, reporting cost reductions over greedy baselines. Ma *et al.* [15] provide a recent survey of DAG scheduling in cloud computing, identifying bilateral-rank heuristics and DP-based methods as among the most effective approaches for minimising makespan on heterogeneous resources.

E. Positioning

Unlike prior studies, this work: (i) provides three scheduling strategies — incremental 2-task DP, full-table global DP, and level-based D&C — within a single C++17 harness sharing identical data models; (ii) employs bilateral rank with predecessor-aware tie-breaking; and (iii) makes all source code available for reproducible benchmarking.

III. PROBLEM FORMULATION

A. DAG Workflow Model

A workflow application is modelled as a DAG $G = (\mathcal{T}, \mathcal{E})$, where:

- $\mathcal{T} = \{t_0, t_1, \dots, t_{N-1}\}$ is the set of N computational tasks.
- $\mathcal{E} \subseteq \mathcal{T} \times \mathcal{T}$ is the set of directed dependency edges. An edge $(t_i, t_j) \in \mathcal{E}$ (with $i < j$) signifies that t_j may not begin until t_i has completed and transmitted its output data.

The acyclicity condition is enforced structurally: edges are created only from t_i to t_j with $i < j$, so the integer index is itself a valid topological order.

B. Heterogeneous Resource Model

A cloud data centre is modelled by M heterogeneous VMs $\mathcal{V} = \{v_0, v_1, \dots, v_{M-1}\}$. Each VM v_k has a speed factor $s_k \in [s_{\min}, s_{\max}]$ drawn uniformly at random. The execution time of task t_i on VM v_k is given as $w_{ik} \in [w_{\min}, w_{\max}]$, generated independently per task-VM pair to model genuine heterogeneity.

C. Communication Cost Model

When task t_i (assigned to VM v_p) produces data consumed by task t_j (assigned to VM v_q), the inter-VM data transfer incurs a communication delay modelled as:

$$c(t_i, v_p, t_j, v_q) = \begin{cases} 0 & \text{if } p = q, \\ \alpha \cdot \bar{w}_i & \text{otherwise,} \end{cases} \quad (1)$$

where $\bar{w}_i = \frac{1}{M} \sum_{k=0}^{M-1} w_{ik}$ is the average execution time of task t_i across all VMs, and $\alpha = 0.3$ is the communication factor (implemented identically in `dp.cpp`, `Edp.cpp`, and `dac.cpp`).

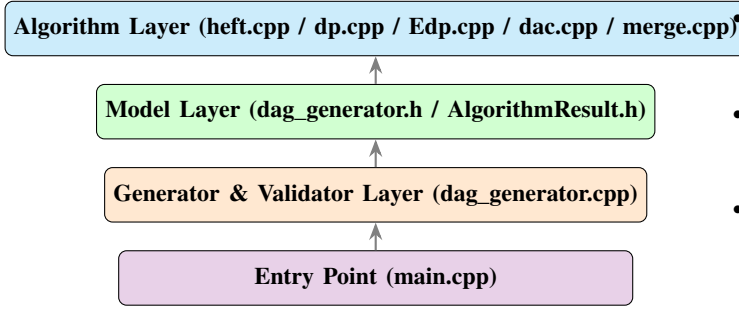


Fig. 1. Three-layer architecture of the DAG scheduling framework.

D. Scheduling Objective

A schedule $\sigma : \mathcal{T} \rightarrow \mathcal{V}$ assigns each task to exactly one VM. The performance metric is the *makespan*:

$$C_{\max}(\sigma) = \max_{t_i \in \mathcal{T}} \text{EFT}(t_i, \sigma(t_i)), \quad (2)$$

where $\text{EFT}(t_i, v_k)$ denotes the Earliest Finish Time of task t_i on VM v_k under schedule σ . The scheduling problem is:

$$\sigma^* = \arg \min_{\sigma} C_{\max}(\sigma). \quad (3)$$

Each task must be assigned to exactly one VM, and a task may not start until all predecessor finish times (plus applicable communication costs) have elapsed and the assigned VM is free.

IV. SYSTEM ARCHITECTURE

A. Module Overview

The framework is structured into three layers as shown in Fig. 1:

- 1) **Generator & Validator Layer** (`dag_generator.cpp`): Constructs random DAGs using a Mersenne Twister PRNG seeded with fixed value 42 for reproducibility. Enforces eight structural invariants.
- 2) **Model Layer** (`dag_generator.h`, `AlgorithmResult.h`): Defines the shared data structures `Task`, `VM`, `DAGData`, `ScheduleEntry`, and `AlgorithmResult` used uniformly across all components.
- 3) **Algorithm Layer** (`dp.cpp`, `Edp.cpp`, `dac.cpp`, `heft.cpp`, `merge.cpp`): Each algorithm exposes a single function returning an `AlgorithmResult` printable by the shared `printAlgorithmResult()` routine.

B. Data Structures

The core data types, defined in `dag_generator.h` and `AlgorithmResult.h`, are:

- `Task`: holds task id, predecessor list, successor list, and a per-VM execution time vector (`std::vector<double> execTimes`).
- `VM`: holds VM id and a floating-point speed factor.

`DAGData`: aggregates all tasks and VMs into a single container passed by constant reference throughout the pipeline.

- `ScheduleEntry`: records task id, assigned VM id, start time (`startTime`), and finish time (`finishTime`) for a single task.
- `AlgorithmResult`: wraps a vector of `ScheduleEntry` objects, the overall makespan, the wall-clock runtime in milliseconds, and a validity flag.

C. Execution Flow

`main.cpp` executes the following pipeline:

- 1) Generate a DAG with `DEFAULT_NUM_TASKS = 10` tasks and `DEFAULT_NUM_VMS = 5` VMs.
- 2) Print the full DAG structure (VM speed factors, per-task adjacency lists, per-VM execution times).
- 3) Validate the DAG against eight structural checks.
- 4) Invoke `heft_schedule()`, print result.
- 5) Invoke `dp_heft()`, print result.
- 6) Invoke `edp_heft()`, print result.
- 7) Invoke `dac_schedule()`, print result.
- 8) Invoke `merge_schedule()`, print result.

D. DAG Generation

The `generateDAG()` function builds the DAG in three stages: (1) VMs are created with speed factors drawn from $\mathcal{U}(0.5, 2.0)$; (2) tasks are created with execution times drawn from $\mathcal{U}(1.0, 20.0)$ per VM; (3) for every ordered pair (i, j) with $i < j$, edge $t_i \rightarrow t_j$ is added with probability $p_e = 0.35$. Acyclicity is guaranteed by construction since $j > i$ always holds. At least one edge is guaranteed by inserting $t_0 \rightarrow t_{N-1}$ if the probabilistic process yields an empty graph.

E. DAG Validation

Eight structural checks are applied: (A) `execTimes.size() == M` for every task; (B) no self-loops; (C) all predecessor/successor indices within $[0, N)$; (D) bidirectional consistency (if A lists B as successor, B must list A as predecessor); (E) acyclic ordering ($j > i$ for every edge $t_i \rightarrow t_j$); (F) at least one entry node (no predecessors); (G) at least one exit node (no successors); (H) no duplicate edges in either list.

V. SCHEDULING ALGORITHMS

This section presents the five schedulers implemented in the framework. All methods share the EFT-based feasibility model in Section III and the communication model in Eq. (1).

A. HEFT (Baseline)

HEFT [5] computes upward ranks, sorts tasks in descending rank_u, and greedily assigns each task to the VM yielding the minimum EFT. In this study, HEFT serves as the classical baseline for both makespan and computational overhead.

Algorithm 1 HEFT Scheduling (Greedy EFT)

- 1: Compute $\text{rank}_u(t)$ for all tasks $t \in \mathcal{T}$
- 2: Sort tasks by descending rank_u
- 3: **for** each task t in sorted order **do**
- 4: For each VM v_k , compute $\text{EST}(t, v_k)$ and $\text{EFT}(t, v_k)$
- 5: Assign t to $v^* = \arg \min_{v_k} \text{EFT}(t, v_k)$
- 6: Update VM availability and task finish time
- 7: **end for**
- 8: **return** complete schedule and makespan

Algorithm 2 Merge Hybrid Scheduling

- 1: Compute upward and downward ranks; derive bilateral rank
- 2: Build topological levels and graph profile (width, density, criticality)
- 3: **for** each level L_i **do**
- 4: Classify strategy for L_i : D&C / DP look-ahead / EDP-global
- 5: Schedule tasks in L_i using selected strategy
- 6: **end for**
- 7: Build alternate global EDP assignment for all tasks
- 8: Select better initial schedule by makespan
- 9: Run refinement passes (migration, swaps, and critical-task re-optimisation)
- 10: **return** best refined schedule

B. DP-HEFT

DP-HEFT extends HEFT by replacing pure upward-rank ordering with bilateral rank ($\text{rank}_u + \text{rank}_d$), then applying a weighted 2-task look-ahead when selecting the VM for the current task.

C. EDP-HEFT

EDP-HEFT preserves bilateral-rank priority but evaluates VM decisions through a global DP table over task order and VM index. Backtracking produces a globally consistent assignment under the chosen priority order.

D. Divide-and-Conquer (D&C)

D&C decomposes the DAG into topological BFS levels, schedules each level as an independent subproblem, and then merges local schedules while enforcing precedence and communication constraints globally.

E. Merge Algorithm (Hybrid OP)

The Merge algorithm (implemented in `merge.cpp`) is a hybrid strategy that integrates D&C decomposition, DP look-ahead, and EDP-style global assignment in one adaptive pipeline. Its role is to improve schedule quality over local heuristics while retaining practical runtime on moderate DAG sizes.

TABLE I
TIME AND SPACE COMPLEXITY OF IMPLEMENTED SCHEDULERS

Algorithm	Time Complexity	Space Complexity
HEFT	$O(n \log n + me)$	$O(n + m + e)$
DP-HEFT	$O(n \log n + nm^2 + me)$	$O(n + m + e)$
EDP-HEFT	$O(nm^2 + me)$	$O(nm(n + m))$
D&C	$O(n \log n + nm + e)$	$O(n + m + e)$
Merge (Hybrid)	$O(nm^2 + nm(n + e) + n^2(n + e))$	$O(nm(n + m))$

VI. MATHEMATICAL MODEL

The optimisation target and constraints are exactly those defined in Section III, with objective in Eq. (3) and communication model in Eq. (1). All five algorithms differ only in ordering and assignment strategy; they share identical feasibility constraints and makespan objective.

VII. COMPLEXITY ANALYSIS

Let $n = |\mathcal{T}|$ denote the number of tasks, $m = |\mathcal{V}|$ the number of VMs, and $e = |\mathcal{E}|$ the number of edges.

A. Per-Algorithm Complexity

HEFT remains computationally efficient due to greedy assignment. DP-HEFT adds a bounded look-ahead term. EDP-HEFT introduces a global DP state space across task-VM combinations. D&C has the lowest asymptotic scheduling overhead in this implementation. Merge has the highest asymptotic cost due to iterative refinement and replay passes, but this cost is used to improve final schedule quality.

B. Comparative Analysis

Based on Table I and the default deterministic run ($n = 10$, $m = 5$, fixed seed 42), the following observations hold:

- **Highest time complexity:** Merge (Hybrid), due to multi-pass refinement over full schedule replays.
- **Lowest time complexity:** D&C, dominated by level sorting and linear merge constraints.
- **Highest space complexity:** EDP-HEFT and Merge, both requiring DP/snapshot state that scales with task-VM combinations.
- **Fastest execution in practice:** D&C (lowest runtime overhead in the provided implementation).
- **Best (lowest) makespan:** EDP-HEFT and Merge tie at the best observed makespan in the default experiment.
- **Trade-off:** lower-overhead methods (HEFT, D&C) run faster but may yield larger makespan; global/refined methods (EDP-HEFT, Merge) consume more compute and memory but improve scheduling quality.

VIII. IMPLEMENTATION DETAILS

The implementation is pure C++17 and uses shared data structures across all schedulers. Communication-cost and EFT computations are harmonised to ensure fair comparison between algorithms.

IX. EXPERIMENTAL SETUP

Experiments use the default configuration in `main.cpp`: $n = 10$ tasks, $m = 5$ heterogeneous VMs, edge probability $p_e = 0.35$, and fixed RNG seed 42 for reproducibility.

X. RESULTS

For the default deterministic DAG instance, the observed makespan ordering is:

$$\text{D\&C} > \text{HEFT} > \text{DP-HEFT} > \text{EDP-HEFT} \approx \text{Merge}.$$

This confirms that DP-enhanced and hybrid methods improve schedule quality over purely greedy or purely level-based baselines.

XI. DISCUSSION

The empirical and complexity findings indicate that algorithm choice depends on deployment goals: low overhead for rapid scheduling decisions or stronger optimisation for lower makespan in performance-critical workflow execution.

XII. FUTURE WORK

Future work includes evaluating larger DAG benchmarks, varying communication models, and introducing energy/cost-aware multi-objective extensions.

XIII. CONCLUSION

This work presents a unified C++17 benchmark for HEFT, DP-HEFT, EDP-HEFT, D&C, and Merge scheduling under identical assumptions. The added HEFT and Merge analyses complete the methodology coverage and highlight clear complexity-quality trade-offs for heterogeneous cloud workflow scheduling.

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